

Rakshith Reddy Aredla

Full-Stack Developer

AI / ML Engineer · RAG Researcher · Toolsmith

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SUMMARY

I build things end-to-end — e-commerce with live inventory and real payments, ML models with measured accuracy, and CLI tools that make other developers faster.

Everything shipped carries a number attached rather than an adjective. 18 public repositories · 99% best model accuracy · 65/65 tests passing · 4 shipped products. accuracy, R^2 and test counts are auditable in a linked repository.

Open to hackathon teams, internships and full-stack / AI roles.

SKILLS

LANGUAGES: Python, TypeScript, JavaScript, SQL, HTML, CSS

FRONTEND & FULL-STACK: React, Next.js, Node.js, React Native, Tailwind CSS, Vite

AI / MACHINE LEARNING: PyTorch, scikit-learn, Naive Bayes, CNN, Random Forest, TF-IDF, Cosine similarity

DATA & BACKEND: Supabase, PostgreSQL, Realtime & Auth, Razorpay, REST APIs, FastAPI, Flask

RESEARCH & OBSERVABILITY: Faithfulness propagation, Cascade detection, Semantic entropy, CUSUM drift monitoring, Root-cause localization

TOOLING: Git, GitHub Actions, Vercel, VS Code, npm, Linux, pip

CURRENTLY EXPLORING: RAG pipelines, Embeddings & vector search, LLM observability

EDUCATION

Gitam University

B.Tech, CSE Core. CGPA 8.2.

2024 – 2028

SELECTED PROJECTS

Arkaira — Online flower shop

Next.js, Supabase, Postgres, Razorpay. A production-style flower e-commerce store built on Next.js and Supabase. Realtime stock that never oversells, Razorpay checkout, and an admin dashboard for the catalogue. Live at: arkaira.vercel.app · Payments: Real · Inventory: Realtime.

<https://github.com/rakshithreddy-aredla/Arkaira> · 2026

Digit Recognition CNN — A convolutional net trained end to end

Python, PyTorch, CNN, Computer Vision. A full PyTorch training pipeline on MNIST — data loading, augmentation, training loop, evaluation — rather than a 40-line notebook that only runs on the author's machine. Test accuracy: ~99%.

<https://github.com/rakshithreddy-aredla/handwritten-digit-cnn> · 2026

rag-pipeline-monitor — Observability for RAG pipelines

Python, FastAPI, RAG, Observability. A research implementation of faithfulness propagation, cascade detection, semantic entropy, CUSUM drift monitors and root-cause localization — built paper-faithfully so the algorithms can be audited against the source. Tests passing: 65/65 · Drift detection: CUSUM.

<https://github.com/rakshithreddy-aredla/rag-pipeline-monitor> · 2026

GitGlance — GitHub profile hackathon-readiness scorer

Python, CLI, GitHub API, Profile analysis. A zero-dependency Python CLI that scores any GitHub profile out of 100 — repos, activity, consistency, project quality — and tells you the exact fixes to raise the number. Transparency over vibes. Dependencies: Zero · Language: Pure Python.

<https://github.com/rakshithreddy-aredla/GitGlance> · 2026

SpamScope — Naive Bayes, no sklearn

Python, Naive Bayes, NLP, From scratch. A spam classifier written from the maths up. Log-space inference, Laplace smoothing and full metrics, implemented by hand so every step is readable and inspectable. Libraries used: None · Inference: Log-space.

<https://github.com/rakshithreddy-aredla/SpamScope> · 2026

EnvGuard — Fail-fast env validation for Node

TypeScript, Node.js, CLI, Validation. Zero-dependency typed environment validation. Declare a schema, get typed config back — and if something is wrong, it lists every error at boot instead of failing at 3am in production.

Dependencies: Zero · Failure mode: Loud & early.

<https://github.com/rakshithreddy-aredla/EnvGuard> · 2026

ACHIEVEMENTS

- Winner — Coogan Hackathon (Unlox), September 2026. Team Leader on a personal lifestyle dashboard.
- Participant — HackWithHyderabad 3.0 (Hack With India x Devnovate), Oct 2026. 8-hour hackathon at Microsoft Office, Hyderabad.

Certificate available on request or at [./certificates/hackwithhyderabad-3.0-participation.pdf](https://github.com/rakshithreddy-aredla/EnvGuard)